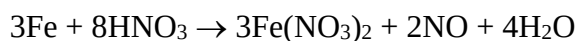


CHEMICAL CALCULATIONS

1. We need to produce 10 moles of PCl_3 by the reaction; $\text{P}_4 + 6\text{PCl}_2 \rightarrow 4\text{PCl}_3$ How many moles of P_4 and Cl_2 are required respectively?

2. What is the mass of oxygen gas, in grams, needed to burn 36 g of carbon ?

3. How many moles of water are produced when 84 g of iron are consumed according to the following reaction ?

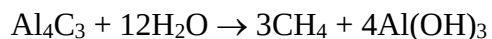


4. What volume of hydrogen gas in liters is produced at STP, if 260 g of zinc completely reacts with sulfuric acid ?

5. How many grams of water can be obtained by heating 49,9 g of copper II sulfate pentahydrate ?

6. What is the mole number of excess reagent when 36 g of pentane (C_5H_{12}) are burned with 112 L of oxygen at STP ?

7. How many liters of CH_4 gas at STP are obtained from the reaction of 3 moles of Al_4C_3 with 108 g of water ? The equation of the reaction is:

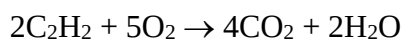
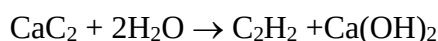


8. After the reaction of a certain mass of nitrogen gas with 12 g of hydrogen gas 4 g of nitrogen remains unreacted. What is the initial mass of nitrogen gas ?

9. How many liters of hydrogen gas at STP are produced at STP when a piece of aluminum reacts with 73 g of 20 % hydrochloric acid ?

10. Heating of 40 g of an impure sample of CaCO_3 yields 6,72 L of CO_2 gas at STP. What is the percentage of CaCO_3 by mass in the sample ?

11. How many liters of oxygen at STP are required to burn the sample of acetylene obtained by the reaction of 12,8 g of calcium carbide with water ?



12. 60 g of C-H-O compound contains 4 g of hydrogen and 32 g of oxygen. What is its empirical formula ?

13. When 2,2 g of a compound containing carbon hydrogen and oxygen is burned 4,4 g of carbon dioxide and 1,8 g of water are produced. What is the empirical formula of the compound ?

14. 0,02 mole of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ solid is heated and its mass decreases by 3,6 g. What is the value of x in the formula ?

15. 10 g mixture of Zn-Cu is reacted with excess HCl and 2.24 L of hydrogen gas are produced at STP. What is the percentage of Cu in the mixture?

16. 25 L mixture of methane (CH_4) and propane (C_3H_8) is burned with 80 L oxygen completely. What is the volume percentage of propane in the mixture?

17. When 41,2 g mixture of Na-Ca is reacted with water 22,4 L of hydrogen gas produce at STP. What is the mass of calcium in the mixture?